Number Theory & Cryptography

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Disclaimer

This document contains the lectures given by Dr. ZAIMI.

Some contents were added as remainders and extras for the students. To separate the contents of the course to actual additions or out of context information, a black band will be added by its side like the one on this comment.

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Chapter 3

Group Actions

Definition 3.1 (Group Actions): *Let G be a group and $X \neq \emptyset$, we say that G acts (operates) on X if there is a homomorphism $\varphi : G \to \mathcal{S}(X)$ the group of permutations of x.*

To avoid complicated notation, we denote $\varphi(g)(x) = \varphi_g(x)$ as $g \cdot x$. Notice in this case $e \cdot x = \varphi(e)(x) = \mathrm{Id}(x) = x$ and $(g_1 g_2) \cdot x = \varphi(g_1 g_2)(x) = \varphi(g_1) \circ \varphi(g_2)(x) = g_1 \cdot (g_2 \cdot x)$.

Definition 3.2 (Group Actions): *Let G be a group and $X \neq \emptyset$ if there is a map $G \times X \to X, (g, x) \mapsto g \cdot x$ where it satisfies the following two identities for any $x \in X$, $g_1, g_2 \in G$, $e \cdot x = x$ and $g_1 g_2 \cdot x = g_1 \cdot (g_2 \cdot x)$ then we say that G acts on X.*

The two previous definitions are equivalent.

Definition 3.3 (Orbit/Stabilizer): *Let G be a group acting on X (X is a G-set) and let $x \in X$, we define the following*

1. *The orbit of x as $O_x = \{g \cdot x \mid g \in G\}$.*
2. *The stabilizer of x as $S_x = \{g \in G \mid g \cdot x = x\}$.*

Proposition 3.4: *S_x is a subgroup of G.*

Proof. By definition 2, we have $e \cdot x = x$ thus $e \in S_x$. Let $g_1, g_2 \in S_x$ then $g_1 \cdot x = x$ and $g_2 \cdot x = x$, so $g_1 g_2 \cdot x = g_1(g_2 \cdot x) = g_1 x = x$ so $g_1 g_2 \in S_x$. Also, $g_1 \cdot x = x \Rightarrow x = e \cdot x = g_1^{-1}(g_1 \cdot x) = g_1^{-1} \cdot x$ thus $g_1^{-1} \in S_x$. ■

Proposition 3.5: *The relation defined on X by $x \mathcal{R} x' \Leftrightarrow x' \in O_x$ is an equivalence relation and the class of x is O_x.*

Proof. Trivial. ■

Definition 3.6:

1. *If $O_x = X$ for some $x \in X$, then we say that G acts transitively on X.*
2. *If $O_x = \{x\}$, then x is said to be stable or fixed, in this case $S_x = G$.*

Proposition 3.7: *Let X be a G-set and let x, y in the same orbit, then S_x and S_y are conjugates.*

Proof. Let $y \in O_x$, then $y = gx$ for some $g \in G$. Let $h \in S_x$, then $hx = x$, $(ghg^{-1})(y) = ghx = gx = y$ therefore $gS_x g^{-1} \subset S_y$ in a similar way we get $g^{-1} S_y g \subset S_x$ by multiplying both sides by g and g^{-1} we get $S_y \subset gS_x g^{-1}$ which gives that $S_y = gS_x g^{-1}$. ■

Proposition 3.8: *Let X be a G-set, then there is a bijection between the set of left cosets of S_x in G and O_x.*

Proof. Let $\mathcal{L}$ be the set of left cosets. Consider the function $\varphi : \mathcal{L} \to O_x, gS_x \mapsto gx$. φ is well defined and injective since $gS_x = hS_x \Leftrightarrow h^{-1}g \in S_x \Leftrightarrow h^{-1}gx = x \Leftrightarrow gx = hx \Leftrightarrow \varphi(gS_x) = \varphi(hS_x)$ and it is surjective by definition. ■

Corollary 3.9: *Let G finite and let X be a G-set then $\# G = \# O_x \cdot \# S_x$.*

Proof. By Lagrange theorem, we have that $\# G = \# S_x [G : S_x]$ and by the previous proposition we have $\# G = \# S_x [G : S_x] = \# S_x \# O_x$. ■

Corollary 3.10 (Class Equation): *Let G be finite and let $O_{x_1}, O_{x_2}, \ldots, O_{x_n}$ be the distinct orbits of G-set X then X is finite and $\# X = \sum_{i=1}^{n} \# O_{x_i}$.*

Definition 3.11 (p-Group): *Let G be a group of order p^n where p is a prime and $n \in \mathbb{N}$, G is said to be a p-group.*

A subgroup of a p-group is also a p-group by Lagrange's theorem.

Corollary 3.12 (Burnside): *The center of a p-group is a p-group.*

Proof. Let G be a p-group, $\mathcal{Z}(G)$ the center of G and consider the action on G defined by $G \times G \to G, (g, x) \mapsto gxg^{-1}$. Let $O_{x_1}, \ldots, O_{x_k}$ be the distinct orbits for this action. Notice that $\forall x \in X, O_x = \{x\} \Leftrightarrow \forall g \in G, gxg^{-1} = x \Leftrightarrow \forall g \in G, gx = xg \Leftrightarrow x \in \mathcal{Z}(G)$. Suppose that $\# G = p^n$, then is either $\# O_{x_i} = 1$ or $\# O_{x_i} = p^{n_i}$ and suppose that $\mathcal{Z}(G) = \{x_1, \ldots, x_s\}$. Thus by the class equation

$$\# G = \# X = \sum_{i=1}^{s} \# O_{x_i} + \sum_{i=s+1}^{n} \# O_{x_i}$$
$$p^n = s + (p^{n_{s+1}} + \cdots + p^{n_k})$$
$$s = p^n - (p^{n_{s+1}} + \cdots + p^{n_k})$$

$s \geq 1$ given that $e \in \mathcal{Z}(G)$ and since p divides the RHS then it divides s so the center is a p-group. ■

Corollary 3.13 (Cauchy): *Let G be a finite group and let p a prime number that divides $\# G$, then there is an element in G of order p.*

Proof. We have already proven this result for the case where G is abelian. Suppose that G is non-abelian. Consider $\# G = n$. For $n = 1$, the result is trivial. Suppose that the result is true for all non-abelian groups of order less than n. If there is a subgroup H of G whose order is divisible by p, we get the result from the induction hypothesis. So suppose that $\# H \not\equiv 0 \bmod p$ for all subgroups H of G with $H \neq G$. Consider the conjugation action $g \cdot x = gxg^{-1}$. We have that $O_x = \{x\} \Leftrightarrow x \in \mathcal{Z}(G)$. Let $O_{x_1}, \ldots, O_{x_r}$ be the orbits of the action where $x_1, \ldots, x_s$ are in $\mathcal{Z}(G)$. Then by class equation we have $\# G = s + \sum_{i=s+1}^{r} O_{x_i}$, on the other hand $\# O_{x_i} = \# G / \# S_{x_i}$, we have that $\# O_{x_i} > 1$ for $x_i \notin \mathcal{Z}(G)$, p does not divide $\# S_{x_i}$ and it divides $\# G$ thus it divides $\# O_{x_i}$, it follows that p divides s and thus $p \mid \# \mathcal{Z}(G)$ which is a contradiction. ■

Theorem 3.14 (First Sylow Theorem): *Let G be a finite group of order $p^n m$ with $n, m \in \mathbb{N}$, p prime and $p \nmid m$, then $\forall k \in [\![1, n]\!]$ there is a subgroup of order p^k.*

Proof. We use induction on k.

- For $k = 1$, the result is true by Cauchy.
- Suppose that the result is true for all $1 \leq k \leq n - 1$. Let H be a subgroup of G with order p^k, we have that $(N(H) : H) \equiv (G : H) \bmod p$ since $(G : H) = p^{n-k} m$ and so $p | (G : H) \Rightarrow p | (N(H) : H)$. Consider $\pi : N(H) \to N(H)/H$, the canonical surjection, $\operatorname{Ker} \pi = H$, $p | \#(N(H)/H)$, by Cauchy's theorem, there exists a subgroup K' of $N(H)/H$ with order p. Then $\pi^{-1}(K')$ is a subgroup K of $N(H)$ containing H. Consider $\pi|_K : K \to K'$, $\operatorname{Ker} \pi|_K = H$, the first isomorphism gives that $K/H \cong K'$ and thus $\# K' = \# H \# K = p^k p = p^{k+1}$ and thus K is a subgroup of G such that p^{k+1}. ■

Definition 3.15: *Let G be a finite group with order $p^n m, n, m \in \mathbb{N}$, p prime and $p \nmid m$, then a subgroup of G with order p^n is called a Sylow p-subgroup of G.*

Notice that the conjugate of a Sylow p-subgroup of G, is also a Sylow p-subgroup.

Theorem 3.16 (Second Sylow Theorem): *All Sylow p-subgroups are conjugates, that is, if P_1 and P_2 have order p^n, then $\exists g \in G, P_2 = gP_1g^{-1}$.*

Proof. Consider the action $P_2 \times \mathcal{L}_{P_1} \to \mathcal{L}_{P_1}, y \cdot gP_1 = (yg)P_1$. Suppose that $O_{gP_1} = \{gP_1\}$ that is $\forall y \in P_2, y \cdot gP_1 = gP_1 \Rightarrow g^{-1}yg \in P_1 \Rightarrow g^{-1}P_1g \subset P_2$ and since $\# P_2 = \# g^{-1}P_2g = \# P_1$ thus $g^{-1}P_1g = P_2$. Hence, if there is a fixed point for this action then P_2 and P_1 are conjugates. On the other hand, the class equation gives

$$m = \frac{mp^n}{p^n} = \frac{\# G}{\# P_1} = \# \mathcal{L}_{P_1} = \sum_{i=1}^{s} \# O_{g_i P_1} + \sum_{i=b+1}^{r} \# O_{g_i P_1}$$

where the first s orbits are of order 1 and the remaining are of order > 1, we have

$$\# O_{g_i P_1} = \frac{\# P_2}{\# S(g_i P_1)} \Rightarrow \forall i \in [\![s+1, \ldots, r]\!], p | \# O_{g_i P_1}$$

Hence $p | \#_{i=s+1}^{r} \# O_{g_i P_1}$ and p does not divide m thus p does not divide $\sum_{i=1}^{s} \# O_{g_i P_1}$ thus $\sum_{i=1}^{s} \# O_{g_i p} \neq 0$. ■

Notation: *We denote the number of Sylow p-subgroups of a group G as n_p.*

Corollary 3.17: *If G is abelian finite and $p \mid \# G$ prime then $n_p = 1$.*

Theorem 3.18 (Third Sylow's Theorem): *Let G be a finite group with order $p^n m$, $n, m \in \mathbb{N}$, p prime and $p \nmid m$, $n_p \equiv 1 \bmod p$ and n_p divides m.*

Proof. We will prove that n_p divides $\# G$. We have that $n_p \equiv 1 \bmod p$. Let φ set of Sylow p-subgroups

$$\begin{aligned} G \times \varphi &\to \varphi \\ (g, P) &\mapsto gPg^{-1} \end{aligned}$$

Let $P \in \varphi$ then $O_p = \varphi$ by theorem 2, then the class equation gives

$$n_p = \# \varphi = \# O_p = (G : S_P)$$

and $\# O_p$ divides $\# G$ then n_p divides $\# G$. ■

Chapter 4

Free Abelian Groups

For an abelian group G, the identity $e = 0$, operation $+$, $\forall n \in \mathbb{N}^*$ define

$$0x = 0$$
$$nx = \underbrace{x + \cdots + x}_{n}$$
$$(-n)x = \underbrace{-x + \cdots + -x}_{n}$$

If $X \subset G$, X generates G if $\forall g \in G, \exists \{x_i\}_{i=1}^n \subset X, \exists \{k_i\}_{i=1}^n \subset \mathbb{Z}, g = k_1x_1 + \cdots + k_nx_n$.

Proposition 4.1: *Let X be a subset of a non-zero abelian group G. Then the following are equivalent:*

1. *$\forall g \in G \setminus \{0\}$, g is uniquely written as a linear combination $g = k_1x_1 + \cdots k_nx_n$ with non-null coefficients $k_i \in \mathbb{Z}^*$ and $x_i \in X$.*
2. *X generates G and $\forall n \in \mathbb{N}^*, \forall x_1, \ldots, x_n \in X, \forall k_1, \cdots, k_n \in \mathbb{Z}, k_1x_1 + \cdots k_nx_n = 0 \Rightarrow k_1 = \cdots = k_n = 0$.*

Proof. $1 \Rightarrow 2$. since $0 = 0x$ we have that X generates G. If $X = \{0\} \Rightarrow G = \{0\}$ which is not the case thus $X \neq \{0\}$. Suppose now $k_1x_1 + \cdots + k_nx_n = 0$, we have then $x_1 = 1x_1 + 0 = 1x_1 + k_1x_1 + \cdots + k_nx_n = (1 + k_1)x_1 + k_1x_1 + \cdots + k_nx_n$, from the unicity of writing we have that necessarily $k_1 + 1 = 1$ and $\forall i \in [\![2, n]\!], k_i = 0$ thus $\forall i \in [\![1, n]\!], k_i = 0$. $2 \Rightarrow 1$. Suppose that for some $g \in G \setminus \{0\}$, we have that $g = k_1x_1 + \cdots + k_nx_n = l_1y_1 + \cdots + l_my_m$, by adding terms with zero coefficients we get that $g = k_1x_1 + \cdots + k_nx_n + k_{n+1}x_{n+1} + \cdots + k_sx_s = m_1x_1 + \cdots + l_nx_n + l_{n+1}x_{n+1} + \cdots + l_sx_s$, hence $0 = g - g = (x_1l_1$

Note: *Undone* ∎

Definition 4.2 (Free Abelian Group): *An abelian group having the property 1 or 2 is said to be free and X is called a basis of G.*

Example:

1. $\mathbb{Z}$ is free with basis $\{1\}$.
2. $\mathbb{Z}^n$ is free with basis $\{e_i = (0, \ldots, 0, 1, 0, \ldots, 0)\}_{i=1}^n$
3. $\mathbb{Z}_n$ is not free, we have that $\forall a \in \mathbb{Z}_n, na = 0$.
4. All finite groups are not free, since by Lagrange, if G is a finite group with order n then $\forall g \in G, ng = e$.
5. $\mathbb{Q}$ is not a free group.

Big Gap Here

Theorem 4.3: *Let G be a free abelian group with rank n. Let $H \neq \{0\}$ subgroup of G. Then, H is a free of rank $s \leq n$. Therefore, there is a basis $\{v_1, \ldots, v_n\} \subset G$ and integers $2 \leq d_1, \ldots, d_s$, $d_1 \mid d_2 \mid \cdots \mid d_s$ and $\{d_1v_1, \ldots, d_sv_s\}$ is a basis of H.*

Theorem 4.4: *Same hypothesis as before, $s = n$ if and only if G/H is finite. Moreover, if $s = n$ $\{x_1, \ldots, x_n\}$ is a basis for G, $\{y_1, \ldots, y_n\}$ a basis for H, $y_i = \sum_{i=1}^n a_{ij}x_j$, then $\# G/H = |\det(a_{ij})|$.*

Proposition 4.5: *Let G be a finitely generated group with n generators, and H a subgroup of G, then H is finitely generated.*

Theorem 4.6: *Let G be a finitely generated abelian group. Then G is a direct product of (possible) finite groups and (possible) free abelian groups. Also, G is a direct product of cyclic groups.*

Proof. Suppose $\{x_1, ..., x_n\}$ generates G, consider the homomorphism

$$\varphi : \mathbb{Z}^n \to G, (k_1, ..., k_n) \mapsto k_1 x_1 + \cdots + k_n x_n$$

take $K = \operatorname{Ker} h$, we consider the cases for K:

- $K = \{0\}$: then $G \cong \mathbb{Z}^n$.
- $K \neq \{0\}$: we have that $G \cong \mathbb{Z}^n / K$ by the first isomorphism theorem, since $\mathbb{Z}^n$ is a free group and K is a subgroup of $\mathbb{Z}^n$ then there is a basis $\{v_1, ..., v_n\}$ of $\mathbb{Z}^n$ and $\{d_1 v_1, ..., d_s v_s\}$ a basis of K with $2 \leq d_1, ..., d_s$ with $d_1 \mid d_2 \mid \cdots \mid d_s$, consider the map $\psi : \mathbb{Z}^n \to \mathbb{Z}^n, k_1 v_1 + ... + k_n v_n \mapsto (k_1, ..., k_n)$ and we have that $\psi(K) = \{(k_1 d_1, ..., k_s d_s, 0, ..., 0) \mid k_1, ..., k_s \in \mathbb{Z}\} = d_1 \mathbb{Z} \times \cdots \times d_s \mathbb{Z} \times \{0\}^{n-s}$, and thus we get

$$\begin{aligned}
\mathbb{Z}^n / K &\cong \frac{\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}}{d_1 \mathbb{Z} \times d_2 \mathbb{Z} \times \cdots \times \mathbb{Z}_s \times \{0\}^{n-s}} \\
&\cong \left(\frac{\mathbb{Z}}{d_1 \mathbb{Z}}\right) \times \cdots \times \left(\frac{\mathbb{Z}}{d_s \mathbb{Z}}\right) \times \mathbb{Z}^{n-s} \\
&\cong \mathbb{Z}_{d_1} \times \cdots \times \cdots \mathbb{Z}_{d_s} \times \mathbb{Z}^{n-s}
\end{aligned}$$

■

Corollary 4.7: *Let $G \neq \{0\}$ be a finite abelian group, then there exists $d_1, ..., d_s \in \mathbb{N}_{\geq 2}, d_1 \mid d_2 \mid \cdots \mid d_s, G \cong \mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_s}$.*